

Low levels of tenofovir in breast milk support breastfeeding in HBV-infected mothers treated with tenofovir disoproxil fumarate.

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Tenofovir disoproxil fumarate (TDF) is recommended for the prevention of mother-to-infant transmission of the hepatitis B virus (HBV). This study investigated the safety of infants whose mothers continued to receive TDF while breastfeeding. Thirty women taking TDF daily from the second or third trimester of pregnancy to three months postpartum were enrolled. Tenofovir (TFV) concentrations in breast milk were determined and compared with those in umbilical cord (UC) blood and amniotic fluid. Infant growth parameters were assessed at birth, and at 3, 6, and 12 months. TFV uptake experiments were conducted in vitro to elucidate the mechanisms of TFV exposure via breast milk. TFV concentrations in breast milk ranged from 1.4 to 11.7 ng/mL within 24 h after dosing in the third month postpartum. The median trough concentration of TFV in breast milk was 3.7 (interquartile range, 2.6-6.2) ng/mL, which is lower than that in UC blood (median = 53.5 ng/mL) and amniotic fluid (median = 531.0 ng/mL). The low permeability of TFV in MCF-10A cells may explain the minimal exposure to TFV in breast milk. Body weights, body lengths, and head circumferences of the breastfed infants were comparable to the national standards for physical development. Infant exposure to TFV from breast milk is much lower than the exposure from placental transfer and swallowing from amniotic fluid. The physical growth parameters of all infants in this study were normal. The findings indicate that breastfeeding is safe for infants of HBV-infected mothers who continue to receive TDF through three months postpartum.

Breast feeding and immunoprophylaxis efficacy of mother-to-child transmission of hepatitis B virus.

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This study was designed to explore if hepatitis B virus (HBV) may be transmitted via breast milk through mother-to-child transmission (MTCT), and assay the immunoprophylaxis efficacy after passive-active immunization. From year 2008 to 2012, 67,720 pregnant women were screened and 1186 HBsAg-carrier mothers and their infants aged 8-12 months were followed in multi-centers of China, among whom HBV markers (HBsAg, HBsAb, HBeAg, HBeAb and HBcAb) and HBV-DNA were measured. HBsAg positive rate of pregnant women was 6.7% (4533/67,720) and infants' immunoprophylaxis failure rate was 3.3% (39/1186). Immunoprophylaxis failure infants were all born to mothers of HBeAg positive and HBV-DNA >6 log10 copies/ml. Among infants of HBeAg positive mothers, HBV infection rate was 9.0% and HBsAg positive rate was 8.3% in breast-feeding group versus 9.2% in formula-feeding group, P=0.761. Occurrence of perinatal HBV infection was indicated in uterus or during delivery. Different feeding patterns had no effects on HBsAb conversion of infants with the implementation of immunization. HBsAg prevalence rate of pregnant women enrolled was 6.7% and immunoprophylaxis failure rate of infants was 3.3%, while the infection rate reached 9.0% in infants of HBeAg positive mothers. Breast feeding did not increase the occurrence of HBV MTCT.

Breastfeeding is not a risk factor for mother-to-child transmission of hepatitis B virus.

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Many clinicians do not encourage breastfeeding in hepatitis B virus (HBV) carriers, since HBV DNA can be detected in breast milk and breast lesions may increase exposure of infants to HBV. The aim of this study was to determine whether breastfeeding may add risk for perinatal HBV transmission. Totally 546 children (1-7-year-old) of 544 HBV-infected mothers were investigated, with 397 breastfed and 149 formula-fed; 137 were born to HBeAg-positive mothers. All children had been vaccinated against hepatitis B but only 53.3% received hepatitis B immune globulin (HBIG). The overall prevalence of HBsAg+, HBsAg-/anti-HBc+, and anti-HBs (≥ 10 mIU/ml) in children was 2.4%, 3.1%, and 71.6% respectively. The HBsAg prevalence in breast- and formula-fed children was 1.5% and 4.7% respectively ($P=0.063$); the difference was likely due to the higher mothers' HBeAg-positive rate in formula-fed group (formula-fed 49.0% vs. breastfed 15.9%, $P<0.001$). Further logistic regression analyses showed that breastfeeding was not associated with the HBV infection in the children, adjusting for the effect of maternal HBeAg status and other factors different between the two groups. Under the recommended prophylaxis, breastfeeding is not a risk factor for mother-to-child transmission of HBV. Therefore, clinicians should encourage HBV-infected mothers to breastfeed their infants.

Breast-Feeding is Not a Risk Factor of Mother-to-Child Transmission of Hepatitis B Virus.

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This study aims at researching the content of hepatitis B virus (HBV) DNA in the breast milk of the mothers carrying HBV and investigating the effects of different feeding methods on mother-to-child transmission (MTCT) of HBV. All infants were voluntarily chosen by their mothers and divided into breast-feeding group and formula-feeding group, which were divided into three subgroups, respectively: HBV-DNA negative (HBV-) group, low viral load (LVL) group and high viral load (HVL) group. HBV load in colostrum and mature milk were both significantly lower than in serum ($P < 0.001$). The positive rate of HBV-DNA in colostrum was positively correlated with HBV load in serum, significantly higher than that of the HBV-Group in colostrum in the LVL Group ($P < 0.05$), and the HVL Group was significantly higher than the LVL Group ($P 0.05$). Breast-feeding is safe for infants with HBV-infected mothers who receive active immunization combined with passive immunization. As well, breast-feeding will neither increase the risks of HBV infection for infants nor weaken their immunity to HBV. However, breast-feeding shall be cautiously applied to pregnant women with high viral load.

Risk of hepatitis B transmission in breast-fed infants of chronic hepatitis B carriers.

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To measure the rate of hepatitis B (HBV) transmission from chronic HBV carriers to breast-fed infants after immunoprophylaxis. Since 1992, information on women with HBV during pregnancy has been collected in a prospective longitudinal study. Those HBV carriers and their infants participating in a county HBV immunoprophylaxis program were identified. Infants were followed for up to 15 months and examined for hepatitis B infection by hepatitis B surface antigen (HBsAg). A total of 369 infants born to women with chronic HBV met the inclusion criteria and received hepatitis B immune globulin at birth and the full course of the hepatitis B vaccine series. We compared 101 breast-fed infants with 268 formula-fed infants. There was no significant difference between the two groups with respect to the number of women who were positive for hepatitis B e antigen (HBeAg) (22% versus 26%, $P=.51$). Three women in the breast-feeding group had liver transaminase abnormalities, compared with six women in

the formula-feeding group ($P = .29$). Overall, there were nine cases of HBV infection transmission (2.4%). None of the 101 breast-fed infants and nine formula-fed infants (3%) were positive for HBsAg after the initial vaccination series ($P = .063$). The mean length of time for breast-feeding was 4.9 months (range 2 weeks to 1 year). With appropriate immunoprophylaxis, including hepatitis B immune globulin and hepatitis B vaccine, breast-feeding of infants of chronic HBV carriers poses no additional risk for the transmission of the hepatitis B virus.

Factors associated with non-compliance with breastfeeding recommendation: a retrospective survey in hepatitis B virus-infected mothers who had taken Nucleos(t)ide analogs during pregnancy.

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We encourage Hepatitis B virus-infected mothers to breastfeed postpartum, even when continuing pregnancy category B nucleos(t)ide analogs (NAs) treatment. However, a large proportion of the Hepatitis B virus-infected mothers were noncompliant with this breastfeeding recommendation. This study aimed to investigate the factors associated with noncompliance with breastfeeding recommendation in Hepatitis B virus-infected mothers who had received NAs treatment during pregnancy. A total of 155 mothers with chronic hepatitis B receiving NAs treatment for preventing mother-to-child transmission during the late gestation period were included and divided into exclusive breastfeeding ($n = 63$), mixed feeding ($n = 34$), and artificial feeding ($n = 58$) groups according to the postpartum feeding methods. Independent variables associated with feeding methods were analyzed using logistic regression analysis. Compared to the breastfeeding and mixed feeding groups, the artificial feeding group had significantly more multiparity, later postpartum timing of stopping NAs treatment, and a lower proportion of having knowledge of NAs medications (all $P < 0.05$). In addition, multivariable logistic regression analysis confirmed that multiparity, later postpartum timing of stopping NAs treatment, and lacking knowledge of medication were independent factors associated with noncompliance with breastfeeding recommendation. Hepatitis B virus-infected mothers who stopped NAs treatment at late postpartum period or had less knowledge of medication were more likely to be noncompliant with breastfeeding recommendation. Strengthening health education for participants taking NAs may be an important method to improve compliance with breastfeeding recommendation.

#38: Hepatitis B in pregnancy screening, treatment, and prevention of vertical transmission.

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Between 800,000-1.4 million people in the United States and more than 240 million people worldwide are infected with hepatitis B virus (HBV). Specific to pregnancy, an estimated prevalence of 0.7-0.9% for chronic hepatitis B infection among pregnant women in the United States has been reported, with >25,000 infants at risk for chronic infection born annually to these women. Vertical transmission of HBV from infected mothers to their fetuses or newborns, either in utero or peripartum, remains a major source of perpetuating the reservoir of chronically infected individuals globally. Universal screening for hepatitis B infection during pregnancy has been recommended for many years. Identification of pregnant women with chronic HBV infection through universal screening has had a major impact in decreasing the risk of

neonatal infection. The purpose of this document is to aid clinicians in counseling their patients regarding perinatal risks and management options available to pregnant women with hepatitis B infection in the absence of coinfection with HIV. We recommend the following: (1) perform routine screening during pregnancy for HBV infection with maternal HBsAg testing (grade 1A); (2) administer hepatitis B vaccine and HBV immunoglobulin within 12 hours of birth to all newborns of HBsAg-positive mothers or those with unknown or undocumented HBsAg status, regardless of whether maternal antiviral therapy has been given during the pregnancy (grade 1A); (3) In pregnant women with HBV infection, we suggest HBV viral load testing in the third trimester (grade 2B); (4) in pregnant women with HBV infection and viral load $>6.8 \log_{10}$ copies/mL, HBV-targeted maternal antiviral therapy should be considered for the purpose of decreasing the risk of intrauterine fetal infection (grade 2B); (5) in pregnant women with HBV infection who are candidates for maternal antiviral therapy, we suggest tenofovir as a first-line agent (grade 2B); (6) we recommend that women with HBV infection be encouraged to breast-feed as long as the infant receives immunoprophylaxis at birth (HBV vaccination and hepatitis B immunoglobulin) (grade 1C); (7) for HBV infected women who have an indication for genetic testing, invasive testing (eg amniocentesis or chorionic villus sampling) may be offered-counseling should include the fact that the risk for maternal-fetal transmission may increase with HBV viral load $>7 \log_{10}$ IU/mL (grade 2C); and (8) we suggest cesarean delivery not be performed for the sole indication for reduction of vertical HBV transmission (grade 2C).

Electroconvulsive Therapy in Refractory and Super-Refractory Status Epilepticus in Adults: A Scoping Review.

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Electroconvulsive therapy (ECT) has been suggested as a treatment option for refractory status epilepticus (RSE) and super-refractory status epilepticus (SRSE). The objective of this scoping review was to conduct an extensive literature review on the role of ECT as a treatment option for RSE and SRSE. We searched Ovid MEDLINE and Scopus for journal articles from database inception until February 2024. Articles were then selected based on predetermined inclusion and exclusion criteria. We identified five retrospective case series with 28 adult patients receiving ECT for RSE or SRSE. ECT was administered within 3-70 days (mean 20 days) after the development of SE, and the mean number of ECT courses ranged from 1 to 12 sessions for each patient. ECT was administered in fixed or titrated doses. A total of 20 out of 28 patients (71%) showed clinical improvement, with two (7%) having complete cessation of seizures. It is essential to note that given the lack of control, there could be overreporting of clinical improvement in these studies. 11 patients (39%) were reported as deceased due to causes that were not directly related to ECT treatment. Four patients (14%) reported adverse effects of ECT, including memory, concentration, and/or cognitive impairment. There are level-4 Oxford Centre for Evidence-Based Medicine evidence and low-level Grading of Recommendations Assessment Development and Education evidence that suggest ECT as a treatment option for RSE and SRSE. In light of the limitations of the existing evidence, clinicians should carefully consider individual patients' clinical contexts when deciding on the appropriateness of ECT as a treatment option. Further research, including prospective studies with controlled designs, is needed to elucidate the efficacy, safety, and optimal regime of ECT in the management of RSE and SRSE.

Electroconvulsive therapy for treatment of intractable seizures. Initial findings in two

children.

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We treated two children with intractable epilepsy with electroconvulsive therapy (ECT) for seizure control. One child showed a change in seizure pattern with treatment, which at greater intensity was also effective in stopping nonconvulsive status epilepticus. The other child showed a decrease in spontaneous seizure frequency during short-term treatment. These findings suggest a possible role for ECT in the management of intractable epilepsy in children who are not candidates for epileptic surgery.

Electroconvulsive therapy for epilepsy and major depression.

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In this first report of electroconvulsive therapy (ECT) for the simultaneous treatment of seizures and depressive episodes, the authors discuss the use of ECT in the treatment of complex-partial seizures and major depression in a geriatric patients who refused antidepressant and antiepileptic medication. ECT has numerous anticonvulsant effects, including elevated seizure threshold and decreased seizure duration, which make it a useful adjunctive therapy in epilepsy that is refractory or not amenable to treatment with medication.
